UNIVERSITY OF CAMBRIDGE

MPhil in Data Intensive Science

Submission: 23:59 on Wednesday 1st April 2026

Coursework Minor A3: Applications of DIS to High Energy Physics
Matt Kenzie and Sven Krippendorf

*This coursework has **two questions**, one on theory and one on experiment. You must answer **both questions**. They carry an equal number of marks.*

*For each question you are exepcted to write a report (no more than 1500 words for each question) along with some corresponding code and plots which address the problem. For the oral assessment you should prepare a short presentation (no more than 8 minutes) on **one of the two topics** and be prepared to answer some questions on it (lasting around 7 minutes). The remaining 15 minutes of the oral will be spent on general questions across both the theory and experiment, with a stronger focus on the aspect that **was not** covered in your presentation.*

This coursework should be submitted via a `GitLab` *repository which will be created for you. Place all of your code and your two reports into this repository. The reports should be in a pdf format and placed in a folder named* `report` *with the names* `report/experiment.pdf` *and* `report/theory.pdf`*, respectively.* **Alongside, you need to submit the .tex sourcefiles and figures.** *You will be provided access to your repository until 23:59 on Wednesday 1st April 2026, after which your work will be deemed ready to assess.*

You are expected to submit code and associated material that demonstrates good software development practices as covered in the Research Computing module.

The two reports should not individually exceed 1500 words (including tables, figure captions and appendices but excluding references); please indicate the word count on the front cover. You are reminded to comply with the requirements given in the Course Handbook regarding the use of, and declaration of use of, autogeneration tools.

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1 Theory: from equivariant neural networks to invariant Lagrangians

In your gitlab repository, you can find the weights of two fully connected neural networks (NN1 and NN2) and some code to load them. The weights were adjusted to obey some symmetry properties approximately. Your tasks are as follows (total marks 21):

1. Show that neural network 1 (NN1) transforms in the fundamental representation of $SU(2)$ and transforms as a singlet under Lorentz transformations. Your answer should include the analytical check and an explicit code implementation testing this equivariance. [4]
2. Show that neural network 2 (NN2) transforms in the adjoint representation of $SU(2)$ and as a spin 1 particle under Lorentz transformations. Your answer should include the analytical check and an explicit code implementation testing this equivariance. [4]
3. For each of these neural network configurations (NN1 and NN2) build a NN that obeys the same equivariance properties. Describe your choices for the networks in your report and argue analytically why they have the appropriate equivariance structure. Then train these neural networks to emulate the output of NN1 and NN2 respectively. [7]
4. Construct a classical Lagrangian involving kinetic terms and a mass term for these fields. Your answer should involve the analytic form and an explicit implementation using the neural networks. Argue - using explicit numerical verification - that your Lagrangian is indeed invariant under $SU(2)$ global symmetry and Lorentz transformations. [6]

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2 Experiment: beauty meson decay analysis

This question is about the strange-beauty meson decay to a pair of oppositely charged kaons and a pair of oppositely charged pions: $B_s^0 \to K^+\pi^-K^-\pi^+$. You are provided with a simulation sample of the signal decay (`mc.root`) and a data sample (`data.root`), which contains a mixture of signal and background decays. The samples can be found here:

`https://cern.ch/mkenzie/mphil/assignment/2526/`

along with an `Info.md` file which describes the structure and features within the files. It is worth noting that the simulation file contains a `weight` branch which corrects for simulation-data discrepancies.

Your task is to identify the sources of background which pollute the sample, isolate the signal decays as best you can and produce projections of the distribution into the different invariant mass combinations of the decay products in order to see if you can determine any resonant structures.

Your report should contain the following sections

1. **Introduction** (~ 250 words)
   Describe the decay process and why it is of interest. You will need to read some of the literature and condense it into a few paragraphs.

2. **Reducing combinatorial background** (~ 400 words)
Train a classifer (of your choosing) which helps to reduce the contamination of combinatorial background (the smoothly falling background from random combinations of $K^+\pi^-K^-\pi^+$). You can isolate a background sample to train against using data events with a B_s^0-candidate mass larger than 5600 MeV$/c^2$. You will need to chose an appropriate set of features and justify the choice in your report.

   You should perform some optimisation of your classifier (trial different networks and optimise the hyperparameters) and report on the result. Your report should include a plot of the classifer output score for the signal and background samples, with an appropriate comparison between training and test samples.

   Find an optimal cut point for the classifier score by making a plot of $S/\sqrt{S+B}$ as a function of the classifer output score, where you make appropriate estimates of both S and B.

   If successful you should see that when placing a cut on the classifer score the smoothly falling combinatorial background component in the data sample is significantly reduced. However several peaking components will still remain.

3. **Reducing other backgrounds** (~ 400 words)
Can you identify what sources these other various peaking backgrounds come from (that are not either a B^0 or B_s^0 meson decaying to $K^+\pi^-K^-\pi^+$). Can you identify any features in the samples that help to reduce these? Can you place or optimise cuts on these features that help to minimise or remove the other backgrounds? You should describe in your report what sources you expect to contribute and justify the choice of cuts that help to reduce them.

4. ***B* candidate mass fit** (~ 200 words)
Perform a fit to the B candidate mass spectrum which should include components for the signal, combinatorial background and any other peaking components you find. Describe what components are included, how you model these components and what parameters you allow to float in the fit (and why).

5. **Projection of the signal component in intermediate masses** (~ 200 words)
Use the result of the mass fit to produce `sweights` for the sample. You should then produce plots of the sWeighted distributions (i.e. signal projection only) for the $(K^+\pi^-)$ invariant mass and the $(K^-\pi^+)$ invariant mass. Can you identify any resonant contributions? Can you explain why there are not only resonant contributions if the background has been removed? Do you see resonances in any of the other 2-body or 3-body mass spectra and would you expect to?

6. **Summary** (~ 50 words)
Briefly summarise your analysis and your findings.

END OF PAPER

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